higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

DATA COMMUNICATION AND NETWORKING
NQF LEVEL 4

27 FEBRUARY 2019

This marking guideline consists of 8 pages.

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SECTION A

QUESTION 1

1.1 A
1.2 D
1.3 B
1.4 C
1.5 B
1.6 B
1.7 D
1.8 C
1.9 B
1.10 D
1.11 B
1.12 B
1.13 A
1.14 B
1.15 C
1.16 C
1.17 B
1.18 A
1.19 D
1.20 B

(20 × 1) **[20]**

QUESTION 2

2.1 True
2.2 True
2.3 False
2.4 True
2.5 True
2.6 False
2.7 True
2.8 False
2.9 False
2.10 True

(10 × 1) **[10]**

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QUESTION 3

3.1 packet
3.2 glass
3.3 MAC/Media access control
3.4 Internet Protocol/IP or IPv4 or IPv6
3.5 beaconing
3.6 Feasibility study
3.7 user utilities
3.8 server sessions
3.9 Network security policy/ security policy
3.10 on-board processor

(10 × 1) **[10]**

QUESTION 4

4.1 F
4.2 H
4.3 J
4.4 G
4.5 A
4.6 I
4.7 D
4.8 C
4.9 B
4.10 E

(10 × 1) **[10]**

TOTAL SECTION A: 50

SECTION B

QUESTION 5

5.1 A – fibre optic
B – wireless (2)

5.2
- First, go to the start menu and into Control Panel.
- Double click on Network and Sharing Centre.
- Click on the 'Change adapter settings' link on the left.
- Right click on your Local Area Connection and left click on Properties.
- Highlight Internet Protocol Version 4 (TCP/IPv4) and click on the Properties button.
- select 'Use the following IP address' and 'Use the following DNS server addresses:'
- Enter in the corresponding settings per their section. If you don't have the settings, or need help with them, call into tech support so we can provide them to you.
- Once the settings have been entered in, click ok on the Internet Protocol Version 4 (TCP/IPv4) Properties window, then click on close on the Local Area Connections properties window. (8)

5.3 Cloud computing means storing and accessing data and programs✓ over the Internet instead of your computer's hard drive.✓ The cloud is just a metaphor for the Internet.✓ Is the delivery of on-demand computing resources or services on a pay-for-use basis✓ (4)

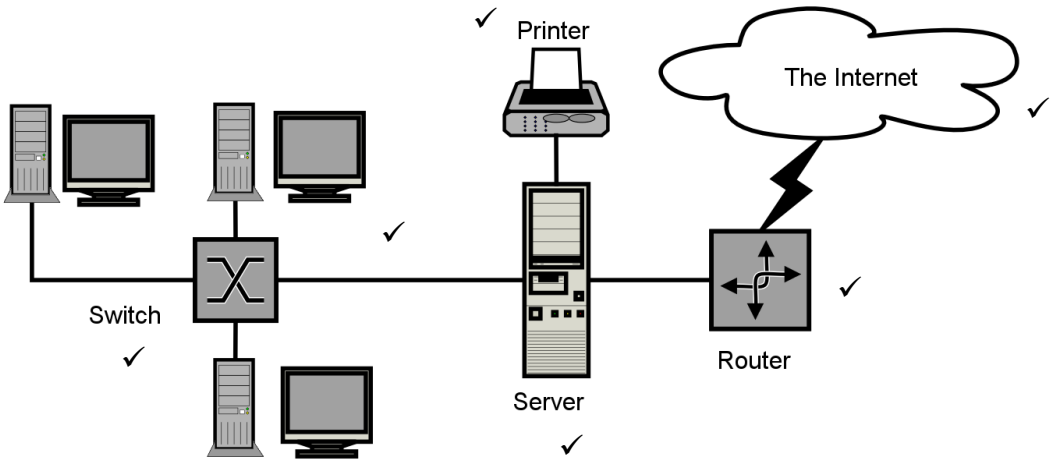
5.4
- Input and output devices, also referred to as 'terminals', provide the starting and stopping points of all communication.
- Telecommunication channels, which transmit and receive data includes various types of cables and wireless radio frequencies.
- Telecommunication processors, which provide several control and support functions.
- Control software, which is responsible for controlling the functionality and activities of the network.
- Messages represent the actual data that is being transmitted.
- Protocols specify how each type of telecommunication systems handles the messages. (6)

[20]

QUESTION 6

6.1 6.1.1 174.24 (1)
6.1.2 208.192 (1)
6.1.3 174.24.0.1 – 174.24.255.254✓ (2)
(Accept any address within the above range.)

6.2



✓ for each component in the diagram
✓ components connected correctly (6)

6.3 Private key encryption uses one key✓ to go both ways✓ whereas Public key encryption uses one key to encrypt ✓ (the public key) and one key to decrypt (the secret key).✓ (4)

6.4 In a client/server network, each workstation is known as a client✓ that can request services from the server.✓ The server is controlled by the network administrator.✓ By looking at the diagram you will see that each client is directly linked to the server.✓ The clients are connected to another by the server.✓ There is centralized administration. ✓ (6)
[20]

QUESTION 7

7.1
- Cut the cable to the length needed. Strip back the cable jacket approximately 1 inch.
- Un-twist each of the 4 pairs and straighten each wire as much as possible between the finger.
- Use the 568-B scheme on one end and the 568A on the other end.
- Bring all of the wires together as closely as possible. Hold the grouped (and sorted) wires together tightly between the thumb, and the forefinger.
- Cut all of the wires at a perfect angle and length from the cable.
- With the connector pins facing up, carefully insert the wires into the connector and crimp. (6)

7.2
- Wire cutter or stripper
- RJ-45 crimp tool
- Cat 5/5e or Cat 6 cable
- 2 RJ-45 plugs
- Cable tester/checker (Any 4 × 1) (4)

7.4
- Measurements of walls and openings to find the correct quantity of cable needed
- Decision on what type of cables to be used based on the required transfer speed, durability and cost
- Checking for suitability of electricity supply for the hardware to be bought
- Arrangement of purchasing of hardware, cabling, connectors and software with the owner's approval
- Scheduling of the installation of LAN to happen on a weekend or any convenient time to avoid disruption on the daily work of the company
- Ensuring that the software and hardware devices are compatible with one another to avoid extra components that could be needed (6)

7.5
- Orientation sessions
- Training sessions
- Job aids
- Technical support information (4)

[20]

QUESTION 8

8.1 The application can run on a network✓ and is able to make use of network resources✓ (2)

8.2
- Size of the network
- Job requirements of the server
- Organisation of storage devices
- File systems to be used
- Identification and naming convention
- Network adapter configuration
- Protocol selection
- Hardware compatibility (8)

8.3 A print queue is a list of documents waiting to be printed.✓ Network administrators configure the server to manage the print queue✓ so that printing requests are prioritised.✓ Printing is prioritised according to size of the print job.✓ Managing a print queue can be done manually.✓ (5)

8.4
- Set up their computers
- Configure their computers
- Customise their computers (3)

8.5
- VPN
- Dial up
- Telnet
- Windows Remote Assistance/Windows Remote Desktop (Any 2 × 1) (2)

[20]

QUESTION 9

9.1
- Save all information
- Close all applications
- Shutdown all workstations using the operating system shutdown procedure
- Switch off all workstations and network printers
- Save information in the applications that are running on the servers
- Close the applications that are running on the server
- Shut down all servers using operating systems' shut down program
- Switch off all servers (Any 7 × 1) (7)

9.2
- User does not know how to access the network
- User physically unable to use the network.
- User does not have access rights to the network
- User has forgotten passwords (Any 3 × 1) (3)

9.3 Data is segmented into packets.✓ Each packet is labelled in sequential order and destination address.✓ Each packet is sent separately.✓ Host receives packet, reads the header, keeps it if it is the intended destination or sends it along.✓ Once destination hosts receive all packets, uses sequential information to reconstruct the data.✓ (5)

9.4 SVC – A path is established when needed and terminated when transmission is complete.✓ Path is maintained only if it's in active use.✓
PVC – Similar to leased lines✓ Pathways are established as permanent logical connections.✓ Pathways exist when not in use.✓ (5)
[20]

TOTAL SECTION B: 100
GRAND TOTAL: 150

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